

Roie Ihia

Electro-Optics M.Sc. Student, Optical Simulation & ML

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M.Sc. student in Electro-Optics (Direct Track, avg. 97) and 4th-year B.Sc. student in Signal Processing & Communication (avg. 92) at HIT, focused on the simulation and validation of optical systems. Research spans on ML-driven optical-device simulation. a first-author Bayesian-optimization pipeline coupled to Lumerical multi-physics solvers and co-authored a real-world optical-sensor data acquisition with ML across an extensive study. Skilled in building automated Python/MATLAB simulation and data-acquisition environments that bridge simulated and measured data.

EDUCATION

M.Sc. Electrical & Electronics Engineering - Electro-Optics at Holon Institute of Technology (HIT) , 2025 -Present. Direct M.Sc. + Thesis track. Cumulative average: 97. Expected graduation: Aug 2027.

B.Sc. Electrical & Electronics Engineering - Signal Processing & Communication at Holon Institute of Technology (HIT), 2022-Present, Cumulative average: 92.5, Expected graduation: Aug 2026.

RESEARCH & PUBLICATIONS

Sample-Efficient Optimization of a Forward-Bias PIN Electro-Optic Phase Shifter - First author (R. Ihia, I. Shalem, D. Malka), manuscript 2026

- Built a closed-loop, ML-driven automated simulation tool coupling multi-physics optical solvers (Lumerical CHARGE / FDE / Interconnect) to a Bayesian-optimization loop through Lumerical's native Python API, an automated simulation tool.
- Raised the valid-design rate from 22% (blind Latin-Hypercube sampling) to 80% (Gaussian-Process surrogate-guided) within a fixed 150-simulation budget and located the optimal design within the system constraint of a silicon MZM phase shifter at 0.649 V·mm modulation efficiency and insertion loss of 15.2 dB/cm.
- Validated the design physically and across the full O-band, extended electro-optic bandwidth from 6.25 to 24.9 GHz with a passive RC equalizer, verified in a 25 Gbps time-domain simulation.

A Lightweight RGB Webcam-Based Gaze Tracking System for Real-World Reading Analysis - Co-author (N. Avital, R. Ihia, et al.), manuscript 2026

- Built an end-to-end pipeline that acquires real-world data from a passive optical sensor across 187 participants under varying illumination, a large-scale experimental data acquisition with no dedicated hardware or GPU.
- Engineered cross-term interaction features and a multi-output XGBoost regressor. reached 3.6° mean angular error, competitive with GPU deep-learning trackers while running real-time on CPU.
- Added a velocity-adaptive Kalman state machine to segment fixations/saccades and export structured behavioral data for downstream analysis.

ADDITIONAL PROJECT

Erasmus+ Blended Intensive Programme - Digital Twin System

Co-developed a proof-of-concept digital twin for real-time device condition monitoring in an international student team, using IoT sensor data and Python to automate device state tracking.

PROFESSIONAL EXPERIENCE

Project Manager at Kinetix AI : 2020 - 2023

Ran sprints, QA cycles, and production embedding for a remote engineering team at an early-stage AI startup; primary technical contact for alpha/beta customers.

Officer (Lieutenant) at Israeli Air Force: 2018 - 2020

Commanded a 10-person unit; planned, managed, and oversaw operations.

Senior Sailing Coach at Ashdod Sailing Club: 2020 - Present

Head coach of a competitive squad; appointed Israeli Youth Team Head Coach (2022).

TECHNICAL SKILLS

Simulation & ML: Bayesian optimization, Gaussian-Process surrogates, XGBoost, Kalman filtering, deep learning, closed-loop simulation pipelines

Optical / RF Design Tools: Lumerical (CHARGE / FDE / FDTD / MODE / Interconnect), CST, HFSS

Programming: Python, MATLAB, C/C++

Lab & Data Acquisition: Oscilloscope, signal/network/spectrum analyzer, signal generator, RF amplifiers; automated Python/MATLAB